ENGR 225 Homework 6 Questions

Assoc. Prof. Clayton Byers

Assigned: 23 Oct 2026

Due: 2 Nov 2026 (11:59pm on Moodle)

Important notes before you begin:

- Note that I've given you an additional weekend to work on this! It will be due on a Monday.
- In case it isn't clear in problem 7, there is a ball joint at F that lies midway between B and D. You should also assume *something* about the connections at A, E, and G.

Problem 1

Determine the force P required to maintain equilibrium in each of the three systems. Assume the block weighs $100\,lb$ (**Ans:** (a) $P = 25\,lb$, (b) $P = 33.3\,lb$, and (c) $P = 11.1\,lb$)

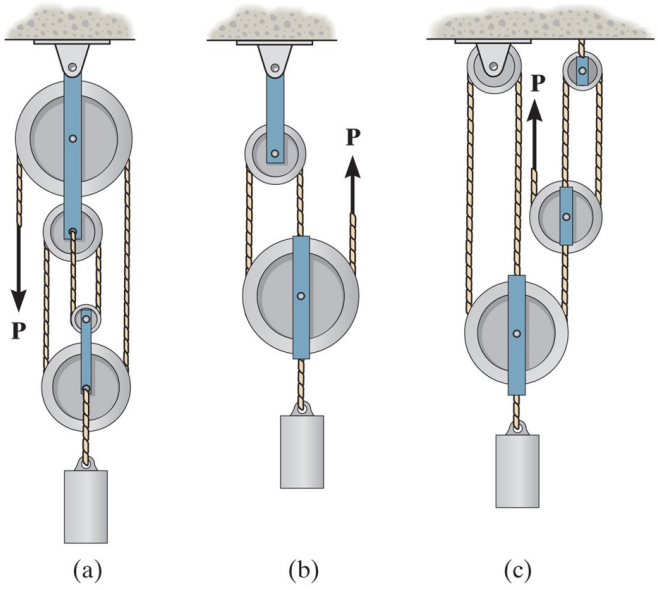


Figure for problem 1.

Problem 2

Determine the reactions at the supports A and B. (**Ans:** $A_x = 2800\,lb$, $A_y = 5100\,lb$, $M_A = 43,200\,lb \cdot ft$, $B_x = -2800\,lb$, and $B_y = 1050\,lb$.)

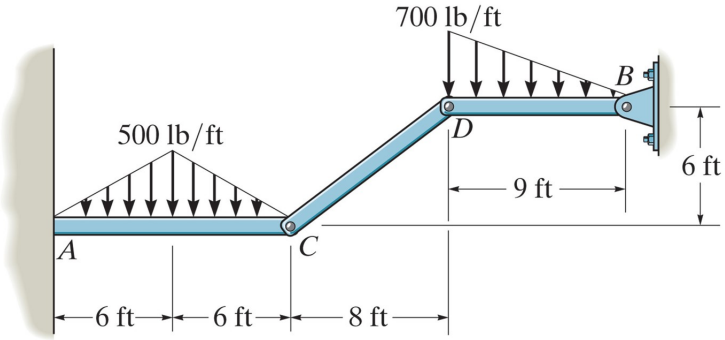


Figure for problem 2.

Problem 3

Determine the horizontal and vertical components of force at pins B and C. Set the mass of the cylinder as $75\,kg$. (**Ans:** $B_x = 1230\,N$, $B_y = 920\,N$, $C_x = -491\,N$, and $C_y = -184\,N$)

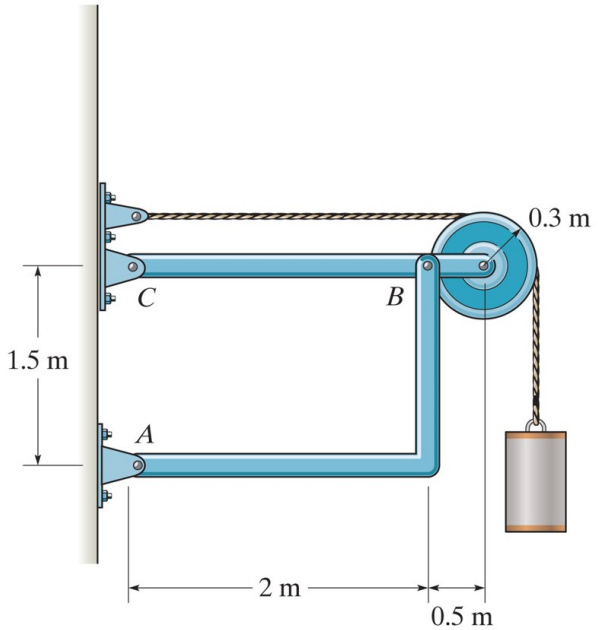


Figure for problem 3.

Problem 4

Determine the force in members FD and BD of the frame. Also, find the horizontal and vertical components of reaction the pin at C exerts on the members ABC and CDE. (**Ans:** $F_{BD} = 25.5\,kN$, $F_{DF} = 20.1\,kN$, Pin C on CDE: $C_x = -18\,kN$, $C_y = -12\,kN$, and Pin C on ABC: $C_x = 18\,kN$, $C_y = 12\,kN$)

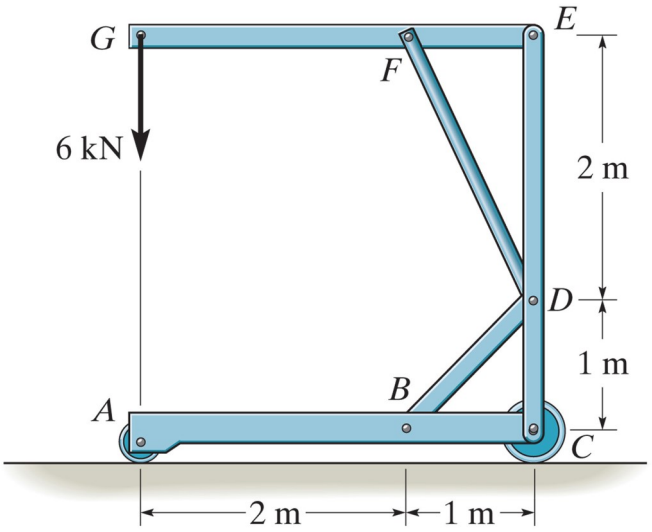


Figure for problem 4.

Problem 5

If a clamping force of $300\,N$ is required at A, determine the amount of force F that must be applied to the handle of the toggle clamp. (**Ans:** $F = 370\,N$)

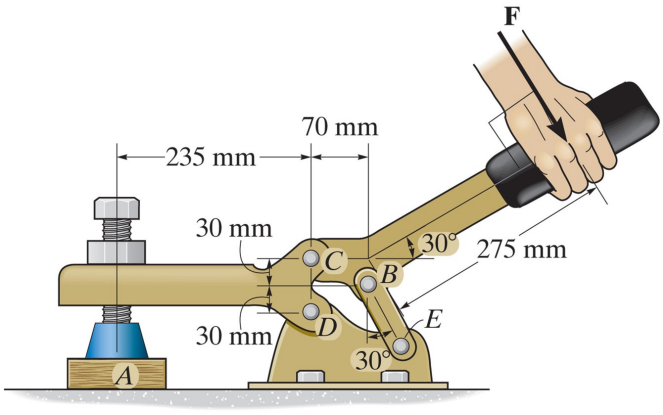


Figure for problem 5.

Problem 6

The piston C moves vertically between the two smooth walls. The spring has a stiffness of $k = 15\,lb/in$ and is unstretched when $\theta = 0°$. Determine the couple moment M that must be applied to AB to hold the mechanism in equilibrium at $\theta = 30°$. (**Ans:** $M = 170\,lb \cdot in$)

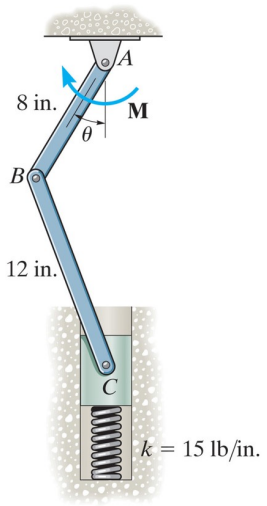


Figure for problem 6.

Problem 7

The four member "A" frame is supported at A and E by smooth collars and a pin at G. All other joints (B, C, D, and F) are ball-and-sockets. If the pin at G will fail when its resultant force is $800\,N$, determine the largest vertical force P that can be supported by the frame. Also, what are the components of force that member BD exerts on members EDC and ABC? (**Ans:** $P = 283\,N$, $B_x = -70.8\,N$, $B_y = -283\,N$, $B_z = -283\,N$, and point D is the same but mirrored in x.)

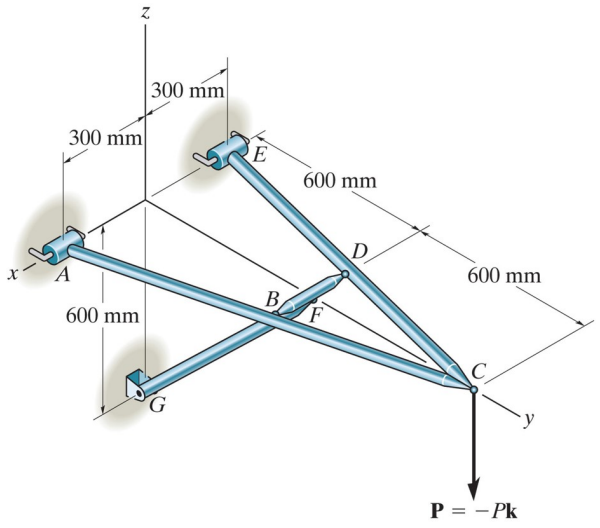


Figure for problem 7.